QUICK REFERENCE • YEAR 1

BIG -O

HOW DOES THE WORK GROW WHEN THE DATA GROWS?

n = number of items



SCAN & LEARN

MENTAL EXPERIMENT

10 → 20

DOUBLE THE INPUT (n).

What happens to the number of operations?

Compare the model values in the cards.

THE SIX PATTERNS TO RECOGNIZE

EASIER → HARDER TO SCALE

O(1)

CONSTANT

Direct access; independent of n.



`a[0]` 1 → 1

O(log n)

LOGARITHMIC

Cut the problem in half.



`n //= 2` 3.3 → 4.3

O(n)

LINEAR

Visit each item once.



`for x in a:` 10 → 20

O(n log n)

N × LOG N

Split, then merge.

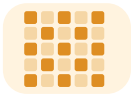


`merge_sort(a)` 33 → 86

O(n²)

QUADRATIC

Compare each with every item.



2 nested loops 100 → 400

O(2ⁿ)

EXPONENTIAL

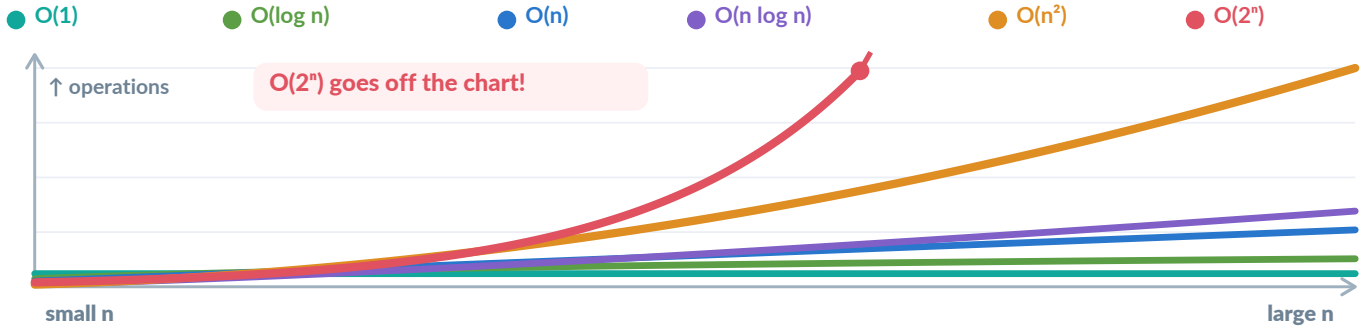
Try every possible subset.



all subsets 1 024 → 1 048 576

GROWTH CURVES: AS n GROWS →

shape matters, not seconds



THREE SURVIVAL RULES

01 LOOPS ONE AFTER ANOTHER

$O(n) + O(n) = O(n)$

add; keep dominant term

02 LOOPS INSIDE LOOPS

$O(n) \times O(n) = O(n^2)$

multiply repetitions

03 DROP CONSTANT FACTORS

$O(3n^2 + 7n + 2) = O(n^2)$

for large n, n^2 dominates

REMEMBER

Big-O = an upper growth bound, not runtime in seconds.

O(1) ≠ exactly one step. Specify the case and resource: time / memory.